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Extended Coiled Tubing Straddle for Integrity Restoration: Three Deployment Cases from the Norwegian Continental Shelf

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Abstract

An innovative remediation technique has been introduced to the Norwegian continental shelf (NCS) leveraging extended-length coiled tubing (CT) straddle systems to isolate corrosion-induced tubing failures. This method restores tubing integrity with minimal intervention effort. Therefore, offering a cost-effective alternative for restoring integrity and production without the need of pulling the upper completion.

A comprehensive suite of downhole tools and surface monitoring systems enabled the deployment of extended coiled tubing straddles exceeding 1,000 m in a single run. The CT string functioned as the straddle body, integrated with a packer and anchoring system. A glass plug was utilized to achieve zonal isolation during installation. The straddle assembly incorporated a gas lift system, a 2-7/8-in seamless CT pipe and an upper packer and anchor configuration. Final reservoir access was restored by hydraulically rupturing the glass plug, completing the installation without requiring conventional workover operations.

The CT straddle system was successfully deployed in three wells on a normally unmanned platform in the NCS. Each well was suffering from tubing integrity failures of up to 1700m corroded tubing, that had halted production. The installations demonstrated that straddle length is governed by well architecture rather than conventional bottomhole assembly deployment constraints. A three-run sequence was executed: (1) installation of the glass plug, packer, and lower anchor; (2) deployment of the stinger, gas lift valve, straddle section, and upper anchor/hanger; and (3) final placement of the stinger and packer to achieve sealing of upper packer and anchor. Well 1 covered 1019 m, well 2 covered 1124 m and well 3 covered 1726.5 m of total straddle length. The straddle system incorporated a gas lift design that restored the functionality of the original completion and allowed future optimization by replacing valves integrated into the lower packer assembly. Reservoir access was restored by hydraulically rupturing the glass plug. This approach proved to be the most cost-effective and operationally efficient method for restoring well integrity and resuming production.

This approach introduces a novel method for restoring integrity and production by integrating a CT pipe as the straddle conduit, anchored with upper and lower packers. Enhanced with a gas lift mandrel, this method enables zonal isolation and lift optimization in a single intervention. It is suited for brownfield

applications where tubing corrosion or mechanical wear is prevalent and can be scaled to an alternative option for water shut off, with minimal operational disruption.

Introduction

Over time, the tubing will naturally degrade until tubing to annulus communication (TTAC) is present. This presents an integrity issue due to the regulatory requirements for well barriers when producing wells, and one will not be allowed to continue producing the well with the integrity issue. As a result, one would like to reinstate integrity and thereby production in a cost-effective manner. Typically, this leads to an evaluation between workover (replacing upper completion) and conventional straddle solutions. The candidate wells require gas-lift for effective production, further restricting the use of conventional straddle solutions. To overcome this challenge, the innovative solution utilizing a straddle system with an integrated gas lift valve, enabling reinstatement of tubing integrity while maintaining the critical gas lift functionality.

Challenge and Problem Definition

Several wells in a mature North Sea oilfield exhibited significant tubing integrity issues, primarily due to multiple leaks caused by pitting corrosion. This corrosion was traced to prolonged exposure to elevated oxygen levels in the gas lift system, where oxygen concentrations exceeded the recommended 5 ppb threshold for 13%Cr martensitic stainless-steel tubing. The corrosion was severe downstream of the gas lift injection point (LSPM), with intensity increasing with depth, affecting intervals extending up to 2000 meters above the injection point.

Given the extensive nature of the damage and the poor project economics, conventional remediation methods such as full workovers and tubing replacement were deemed unfeasible. A cost-effective and technically robust solution was therefore required—one that could:

- Cover long, degraded tubing intervals spanning several thousand meters,
- Maintain access for future interventions,
- Enable deep gas lift injection to enhance production,
- Comply with stringent well control and barrier requirements during both installation and operation.

Expanded Methodology

The methodology faced multiple challenges during the planning phase related to:

- Well barrier status while installing the straddle system.
- Maintaining the ability to use gas lift system post intervention.
- String compatibility with future WL(Wireline)/SLK(Slickline) intervention inside or through the CT straddle system. This is required in case of GLV change-out or future plug & abandonment (P&A) scope.

Well barriers were maintained by utilizing a glass plug and packer system installed in the first CT run to create a barrier and foundation for the subsequent runs. Thereafter, the tubing and annulus volume above the glass plug was filled with inhibited seawater and monitored at surface utilizing trip tank to ensure the liquid level remained stable and the barrier integrity could be monitored.

Utilizing an innovative BHA solution for the lower straddle BHA, incorporating a gas lift valve meant the existing gas lift system can still be used for enhancing production. However, it permits one gas lift valve

into the straddle installation. Depending on the existing configuration and straddle depth this could lead to a reduction of the number of existing gas lift valves.

CT String design

CT string design was optimized to ensure future access through the straddle system. The string was specially ordered with regards to grade, wall thickness and internal roughness. The internal weld seam resulting from the manufacturing process was removed, to minimize future wear on any bottom hole assemblies (BHA) to pass through. Seamless internal surface was proven necessary through yard trials, where a SIT was performed to simulate BHA wear as its run through a "normal" 2-7/8 string. String grade was selected based on material compatibility with well fluids while maintaining sufficient mechanical properties, grade 90-string came out as a suitable option balancing mechanical properties and resistivity towards the well environment.



Figure 1—A system integration test was conducted to see if subsequent interventions could be performed through the CT straddle. During those tests it was found out that the internal seam weld of the CT pipe would damage rubber seal area of plugs or other intervention tools. The picture displays the damage seen on one of the plugs run though the CT pipe. As an outcome the CT straddle used had the internal seam weld removed.

Straddle System Design

To address these challenges, a CT Straddle System was selected as the intervention method. This approach enabled targeted isolation of leaking tubing sections without necessitating full tubing replacement. The methodology was developed collaboratively by engineers from different service companies and the operator, drawing on prior field experience and adapting to the unique conditions of each well.

Key Elements of Methodology

CT Straddle Installation. Each well received a custom-designed CT straddle assembly incorporating packers and a gas lift mandrel with a pre-installed gas lift valve (GLV). These assemblies were engineered to span the corroded tubing zones and restore mechanical integrity while enabling continued production.

Glass Plug Integration. To ensure well control during installation, V0-qualified glass plugs were integrated into the lower packer assemblies. These plugs provided a reliable barrier, allowing pressure testing of the installation prior to re-establishing communication with the reservoir.

Operational Efficiency. The interventions were executed as part of a coordinated campaign to minimize mobilization costs and maximize learning across wells. Tight time and cost controls were maintained throughout the operation.

CT String Selection. The CT string was carefully selected to meet installation and flow requirements while also allowing for future interventions. This dual-purpose design ensured long-term flexibility and operational resilience.

General installation considerations

The straddle system design leveraged industry-standard practices for installing CT straddles and velocity strings, commonly used to restore well integrity or deliver gas lift deeper than the original gas lift mandrels. The installation typically involved two or three runs, depending on the tailpipe weight.

1. Lower Packer Installation: The first run deployed the Lower Packer, which included a seal bore and an integrated glass plug for initial isolation.
2. Straddle Assembly Deployment: The second run installed the tailpipe (straddle) assembly, anchored using either an upper packer or a mechanical hanger system, selected based on the tailpipe weight.

Given the high hang-off loads in this campaign—some exceeding 10,000 lbs—a mechanical hanger system was used to transfer the tailpipe load to the production tubing. This avoided overstressing the packer elements and ensured reliable sealing. The upper packer was then installed above the hanger to provide hydraulic sealing capability.

To enhance production, gas lift modeling confirmed the benefit of injecting gas deeper in the wellbore. A gas lift mandrel was placed directly above the Lower Packer, enabling gas lift injection at the base of the straddle. The mandrel design allowed for GLV replacement without retrieving the entire straddle, improving long-term serviceability.

For system recovery at the end of the well's life, a QN Nipple was included to accommodate a QX lock mandrel. All components were qualified to ISO 14310-V0, ensuring compliance with well barrier standards.

Installation procedure

After reviewing the well status and barrier management, it was concluded to perform the installation in 4 runs plus an additional surface leak testing activity. Therefore, the installation consisted of 5 activities. A detailed description of each operational step is outlined followed by a visual description of the installation steps.

Run #1 – Drift & Cleaning setting areas. The objective of Intervention Run #1 was to ensure unobstructed access to the wellbore and to clean the designated packer setting areas, thereby increasing the likelihood of success for the straddle installation. The BHA used for this operation consisted of a rotating jet cleaning configuration equipped with a drift ring. This setup allowed for both jetting in case of obstructions and effective removal of debris and deposits from the tubing, particularly in the critical areas designated for packer and CT hanger installation.

The cleaning focused on three key intervals: the lower packer setting area, the upper packer and CT hanger setting area, and the DHSV region. These zones were targeted to remove any debris or deposits that could interfere with tool setting or sealing.

In the event that obstructions could not be removed using the jetting tool, the contingency plan was to make up and run in hole with a 3.75" OD MDX mill and a motor.

This preparatory run was essential for confirming access and readiness of the wellbore for the installation of the straddle system, ensuring that subsequent runs could proceed with minimal risk of mechanical interference or setting failure.

Run #2 – Installing lower packer and glass plug. The objective of Intervention Run #2 was to install the lower packer assembly and glass plug at a pre-determined depth, establishing the lowermost isolation point for the straddle system. The packer assembly included a 1.5" QN nipple to facilitate future straddle retrieval and the glass plug serving as both a fluid loss barrier and a means of leak testing the straddle after installation. The installation was executed using a hydraulic setting tool, equipped with a 0.312" minimum ID ball seat and a 0.5" setting ball, following standard procedures.

After setting the packer, the tubing and annulus volumes above the glass plug were displaced with inhibited seawater (ISW) and MEG-mix to allow surface monitoring of the packer and plug integrity. Displacement was performed by pumping down the A-annulus and taking returns to production via the choke and donor well, with a maximum pump rate of 1.5 bpm (239 lpm) due to the 20/64" orifice gas lift valve still installed in the LSPM. Gas was bled off from both the A-annulus and tubing prior to displacement, and lube and bleed procedures were applied to ensure complete removal of gas. The B-annulus was monitored continuously during testing, with pressure bled off if it exceeded expected ballooning. Following displacement, the well was leak tested to a maximum pressure of 149 bar against the glass plug and packer.

It was noted that the seamless GT90 2-7/8" CT string used in this run had a lower collapse rating than standard—244 bar—which was a critical consideration for pressure and leak testing activities. A contingency plan was in place: if the packer could not be set, the assembly would be pulled out of hole (POOH), inspected, and a backup assembly prepared for installation at an alternative depth. Similarly, if the leak test failed, the lower packer assembly would be retrieved and replaced.

Run #3 – Running lower straddle BHA, straddle string and upper straddle BHA. The objective of Intervention Run #3 was to install the straddle string and gas lift mandrel. This involved cutting the CT string to the required length, stinging into the previously installed Lower packer, and hanging off the straddle string using a CT hanger. The operation began by connecting the AnchorSeal to the CT string via a connector. Proper spacing was ensured so that the CT hanger would be positioned at correct depth when the anchor seal latched onto the lower packer.

Once the lower BHA was deployed and the desired straddle length was run in hole, the well's stability was verified by monitoring inflow and losses via the trip tank. Following confirmation of well stability, the CT string was cut above the CT BOP, and the upper BHA was deployed. The CT hanger was then installed and set using a 4" GS flow release running tool in compression mode. The setting procedure involved picking up the CT string approximately 0.3 meters, setting it back down, and then picking up again to engage the hanger. Releasing the hanger was achieved by simply picking up the CT string, which activated the automatic J-slot mechanism, allowing repositioning or retrieval.

An exit strategy was defined to manage potential risks. If a loss of hydrostatic head or hydrocarbon influx occurred while running in hole, the plan was to pull out of hole (POOH) while maintaining fluid level. If such an event occurred after containment was broken (i.e., after cutting the CT string at surface), the well would be secured by dropping or cutting the CT string and closing in the well. These procedures were supported by a detailed risk assessment.

Run #4 – Running upper packer. The objective of Intervention Run #4 was to install the upper packer at the designated depth, completing the upper isolation point of the straddle system. The packer was deployed using a hydraulic setting tool, in accordance with standard procedures. To facilitate reliable activation, a 0.5" setting ball was pre-installed into the tool, as other runs had shown difficulty in pumping the ball over the gooseneck. Additionally, a 5 mm orifice sub was included above the ball seat sub to prevent trapped pressure between check valves and the ball.

The installation procedure involved stinging into the seal assembly and setting the upper packer hydraulically. The packer was engaged by activating the hydraulic setting tool via the conventional ball drop method. Depth control was managed using the coiled tubing depth counter, and care was taken to ensure the pipe and slip ram inserts were in good condition prior to cutting the string above the CT BOP.

This run marked the final step in establishing the straddle isolation envelope, enabling zonal integrity restoration while maintaining gas lift functionality. The successful installation of the upper packer ensured that the straddle system was securely anchored and sealed, ready for leak testing and subsequent production.

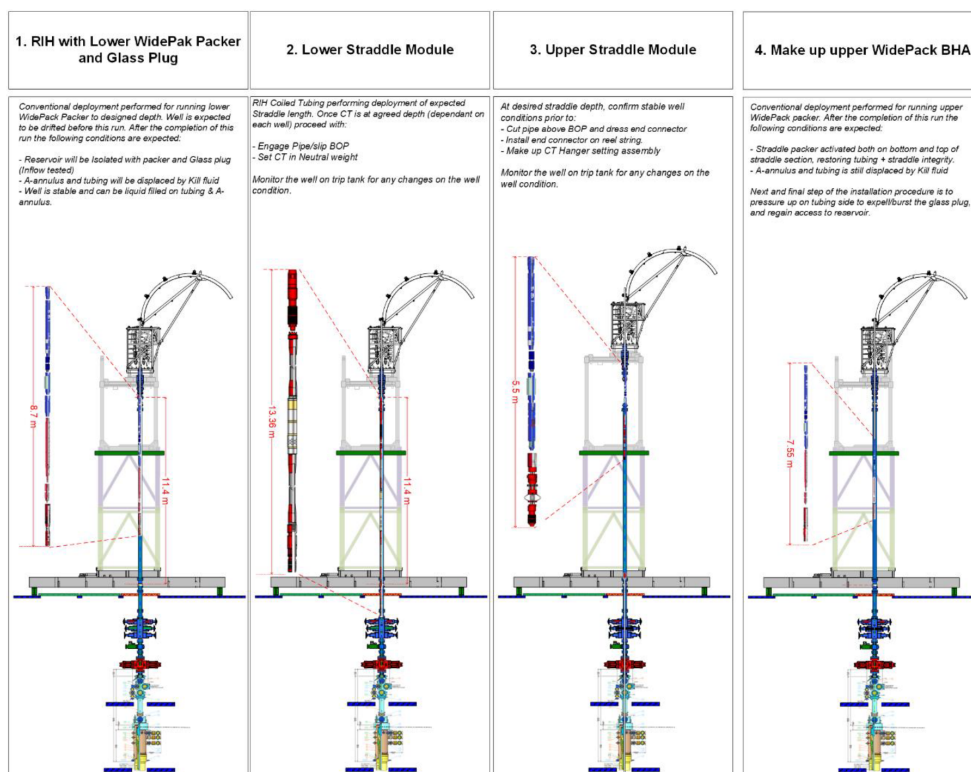
Activity #5 Leak test straddle assembly. Objective of run#5 is to verify the integrity of the installed gas lift straddle system through pressure leak testing. The leak test was performed with consideration of the CT

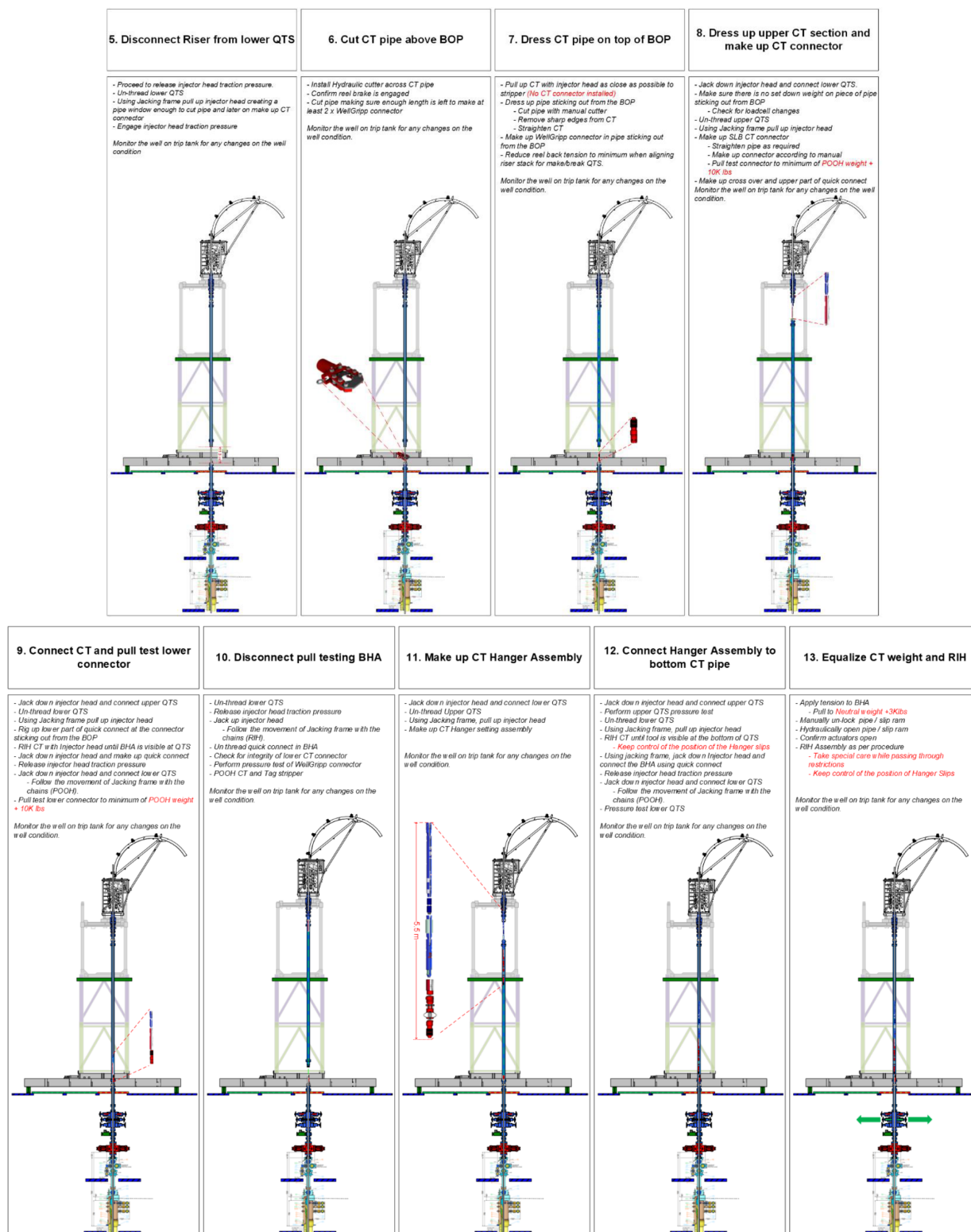
string's reduced collapse rating. A pressure test matrix was developed to evaluate safety margins from 90% to 110% of the target pressure, all of which passed under the collapse threshold. The seamless GT90 string's lower collapse rating was a critical constraint during pressure testing.

Exit Strategy (Contingency Plan): In case of a failed leak test, troubleshooting would be initiated. If integrity could not be confirmed, the well would be abandoned without opening the deep-set glass plug to maintain reservoir isolation.

After the straddle system was successfully leak-tested the glass plug was pressurized and burst to allow access to the reservoir.

Visualization of surface steps for installation





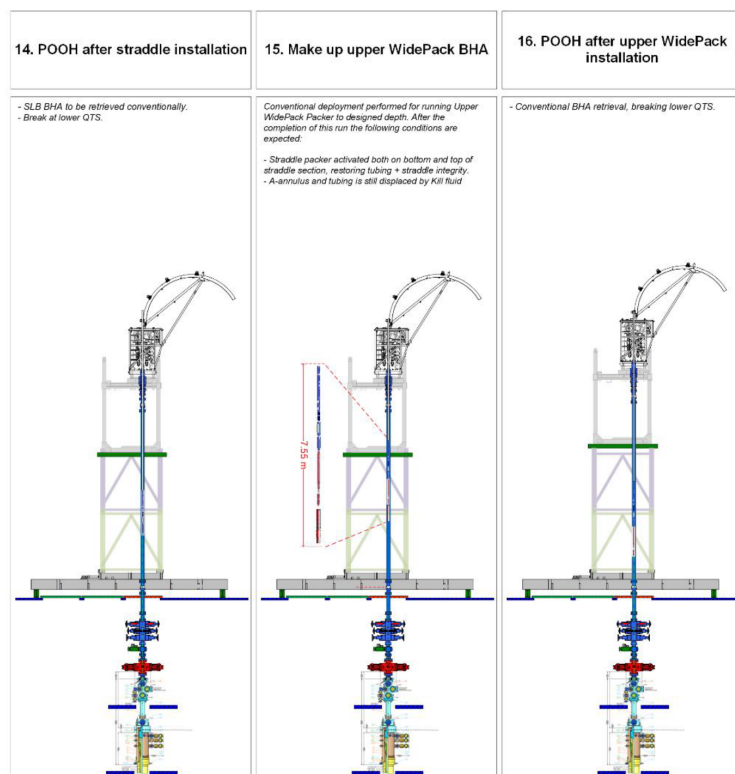


Figure 2—The 16 parts for the installation are displayed visually for each critical step at surface

Well barrier comparison before and after

Figure 3 is a schematic of the entire straddle system after installation. It details the upper packer assembly the upper hanger assembly the CT pipe, the gas lift mandrel, the lower anchor and packer and the glass plug for reference.

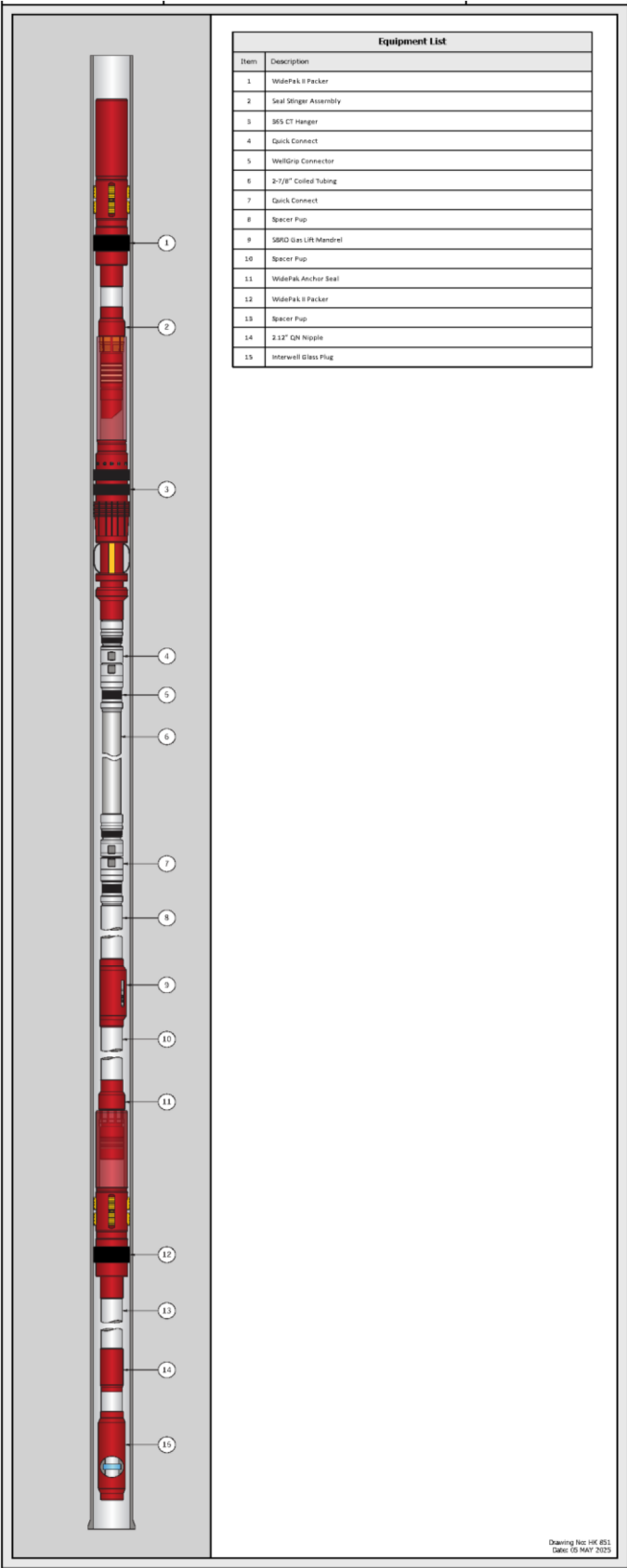


Figure 3—Schematic of the entire straddle system as placed in the well.

Well barrier comparison before and after

The overall objective of the intervention is to re-instate the tubing integrity by means of installing the straddle, while retaining the gas lift functionality. This can be illustrated as shown in Figure 2 below, where the well barrier schematic (WBS) before and after the intervention is shown. Areas of particular interest are highlighted in yellow where the area of the sealing of the upper and lower packer and the CT straddle is highlighted.

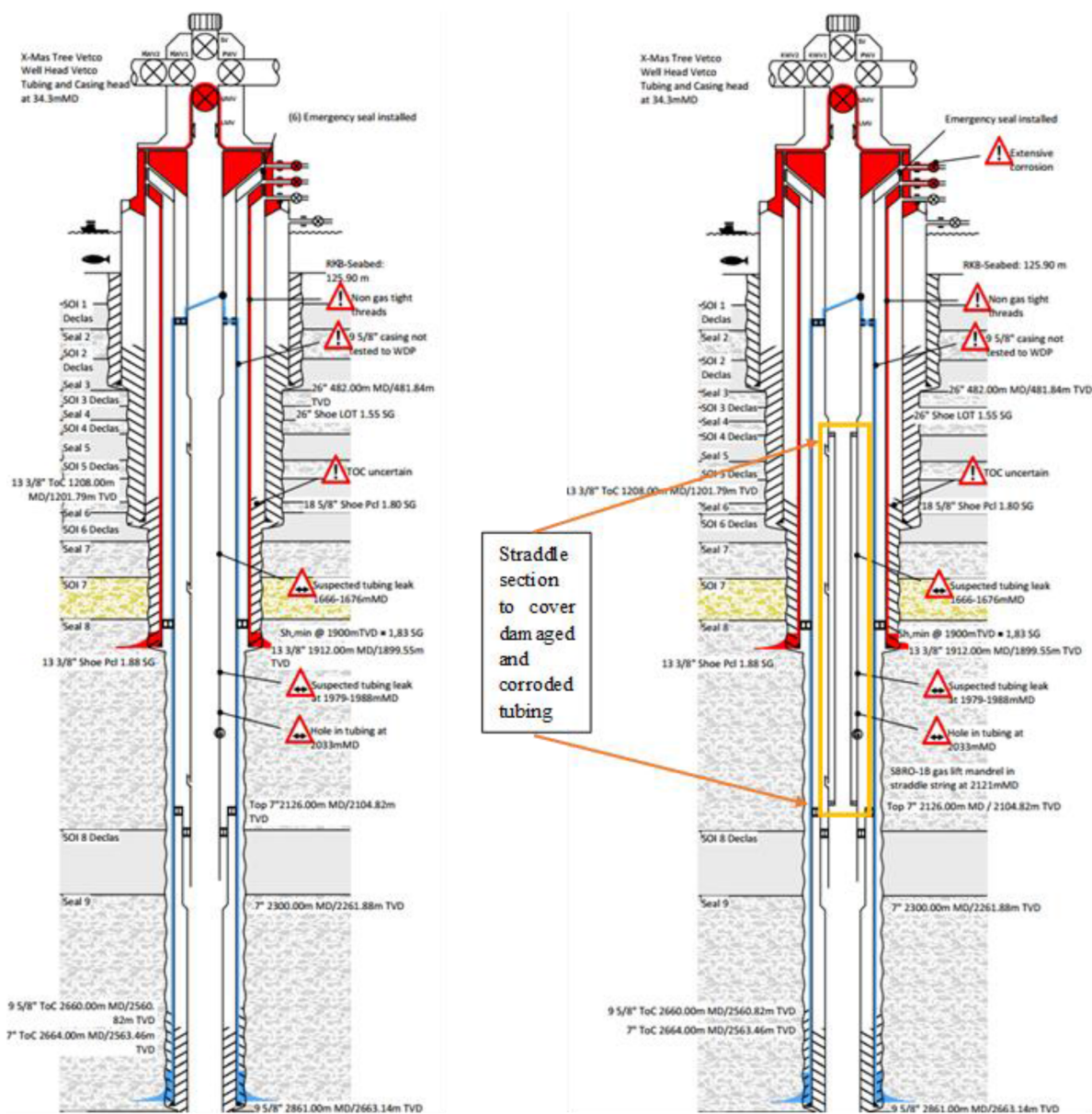


Figure 4—The well barrier status before and after the installation highlighting the straddle section.

Results and discussion

The campaign yielded results as shown in Table 1. A total straddle length of 4018 meters were installed in the three wells, consistently ahead of the estimated schedule, lower than expected cost and higher than forecasted production for the 2 out of 3 producing wells.

Table 1—This table shows the results of the three wells and the installation.

	Well #1	Well #2	Well #3
Straddle Distance (meters)	1 127	1 149	1 743
Objective Score (%)	100%	100%	0%
Actual Duration (days)	22.17	13.8	14.29
P10 Duration (days)	23.1	17.7	12.2
Actual Inc Oil Rate (mboe/day)	0.37	0.40	-

Challenges Well 1

The first well was a sub-hydrostatic oil producer, first put on production in 2006. Gas lift were installed with 2 IPO's at the upper side-pocket mandrel and middle side-pocket mandrel, and an orifice valve at the lower side-pocket mandrel. The straddle technique worked as intended, with the main challenge faced being the ball dropping of 1/2-inch balls in a sub-hydrostatic well inside a 2-7/8" string without pumping excessive rates. It resulted in the lower WidePak being set hydraulically on differential pressure rather than ball dropping. Apart from the ball dropping, there were no further challenges relating to the straddle installation faced on the first well. Figure 3 provides the acquisition for the whole installation on the first well and it was completed in approximately 6 operational days.

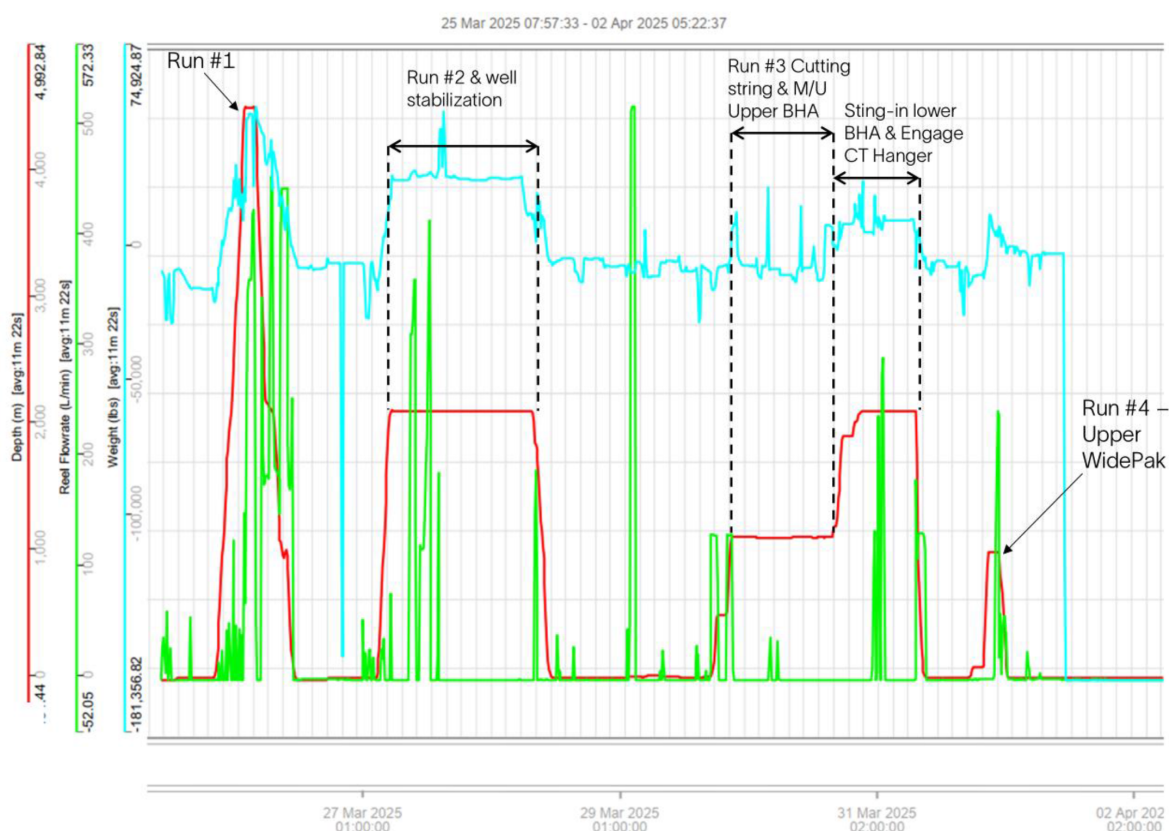


Figure 5—This picture shows the acquisition plot for the straddle installation on well 1. The parameters for the depth (red), the running weight (light blue) and the pump rate (green) are displayed

Challenges Well 2

The second well incorporated learnings from the first one regarding the ball dropping, which was resolved by changing ball material to aluminum, which is lighter and therefore easier to pump. Performing deck tests

confirmed rate required to launch the ball. It was, however, more difficult to stabilize the second well due to volume in the a-annulus located between the production packer and lowermost connection point between tubing and a-annulus being difficult to circulate. As seen in Figure 4 there is a significant difference between well #1 and well #2 on the stabilization period. The rest of the installation went as planned and no other observations were made.

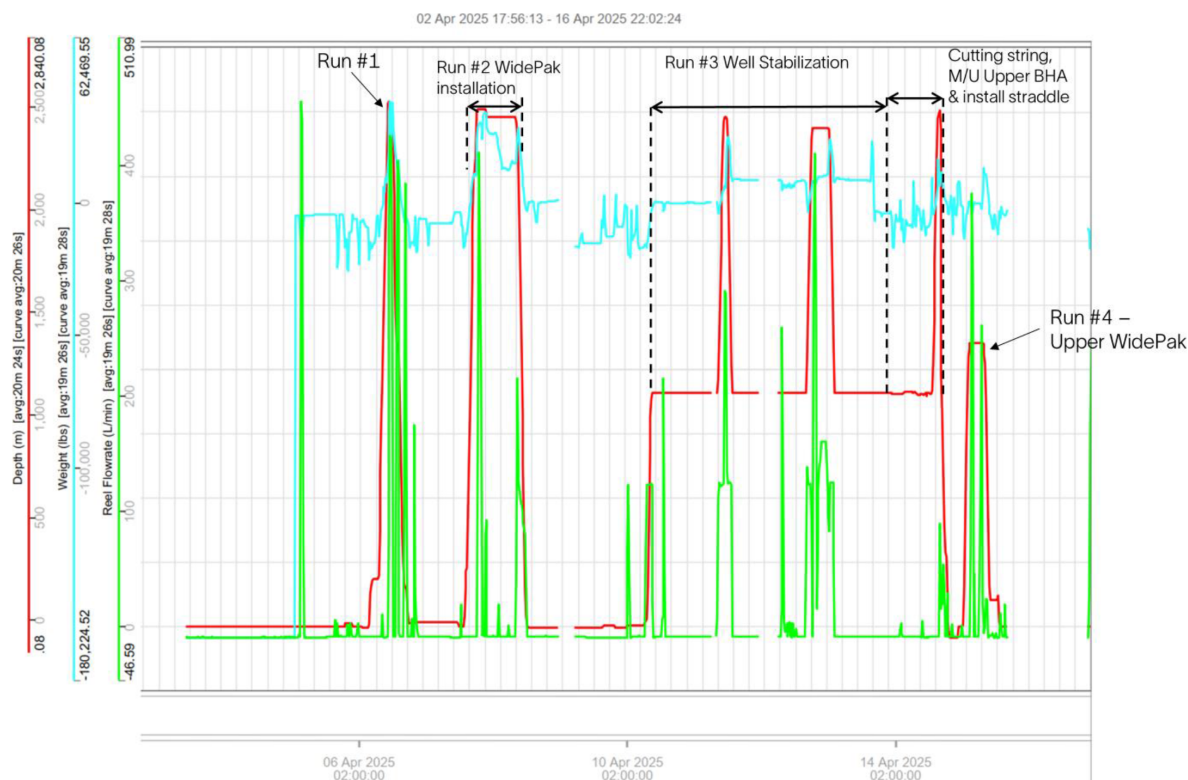


Figure 6—This picture shows the acquisition of plot of straddle installation on well 2. The parameters for the depth (red), the running weight (light blue) and the pump rate (green) are displayed. It is noted that the well stabilization took longer than on well number 1.

Challenges Well 3

The third well incorporated learnings from both the previous ones on both the ball dropping and launching and also well stabilization. This resulted in a very efficient operation successfully installing the straddle system and pressure testing it to required value. However, during the final pressurizing to expel/burst the glass plug a leak/communication was established between the tubing and a-annulus. This led to quick depressurization. Subsequent analysis revealed the leak to be substantial. A WL campaign has been planned for Q3-2025 to further investigate the root cause.

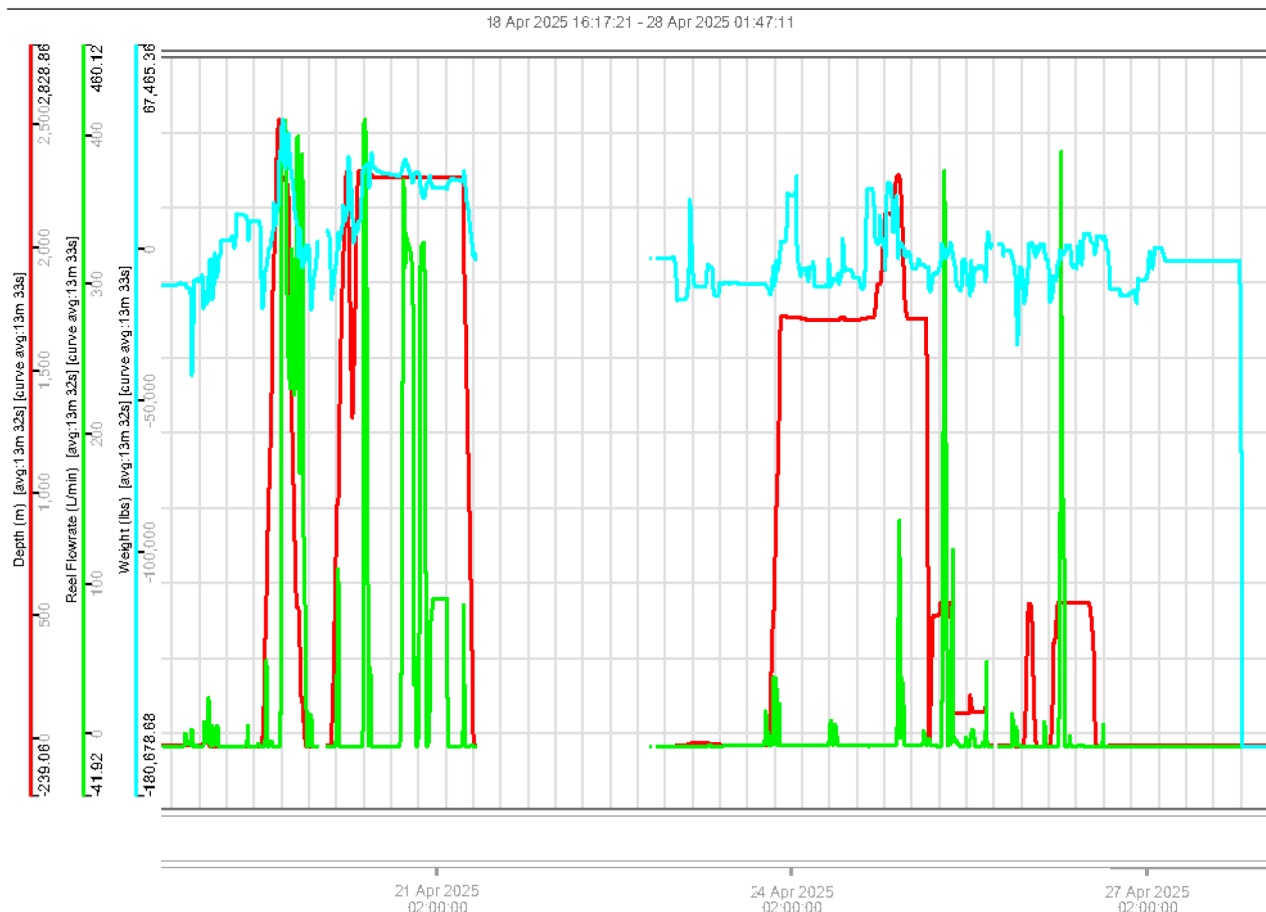


Figure 7—This picture shows the acquisition of plot of straddle installation on well 3. The parameters for the depth (red), the running weight(light blue) and the pump rate (green) are displayed.

Conclusion

The extended coiled tubing (CT) straddle system has proven to be a reliable and cost-effective solution for restoring tubing integrity in mature wells suffering from corrosion-induced failures. The successful deployment across three wells on the Norwegian Continental Shelf demonstrated that long straddle installations—exceeding 1,000 meters—can be achieved without conventional workover operations, even in wells with complex geometries and challenging pressure regimes.

Operational Efficiency: The multi-run installation strategy enabled precise placement of packers and straddle components, with each well completed ahead of schedule and below budget.

Integrity Restoration: The integration of V0-qualified glass plugs and optimized CT string design ensured robust zonal isolation and well control throughout the intervention.

Production Enhancement: The restored wells delivered incremental oil production of up to 340 boe/d, validating the effectiveness of the straddle system in re-establishing flow and gas lift functionality.

Gas Lift Integration: Incorporating a gas lift mandrel directly into the straddle system allowed for deep gas lift injection and preserved artificial lift capability without compromising well integrity. This integration enabled continued production and future optimization through valve replacement, all without retrieving the straddle.

Design Adaptability: The CT string was custom-engineered to support future interventions, with internal seam welds removed to prevent tool damage—an improvement confirmed through system integration testing.

Lessons Learned: Challenges such as ball drop reliability and well stabilization were addressed through iterative improvements.

This approach by combination of long CT straddle system together with the gas lift mandrel offers a scalable solution for brownfield applications, particularly where tubing degradation or mechanical wear limits conventional remediation. It also opens pathways for future interventions, while maintaining the gas lift potential.

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